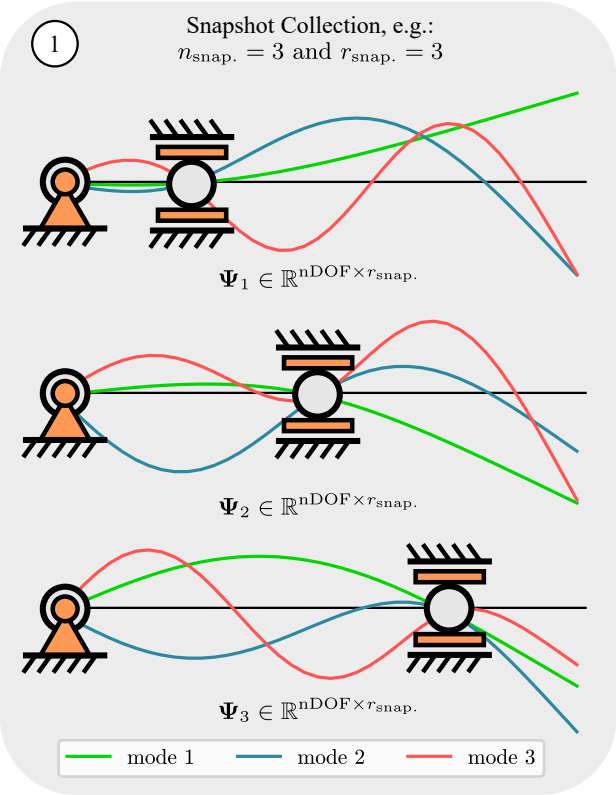


2 Assemble *Pool*

$$\boldsymbol{\Psi}_{\text{all}} = [\boldsymbol{\Psi}_1, \boldsymbol{\Psi}_2, \boldsymbol{\Psi}_3]$$

$$\boldsymbol{\Psi}_{\text{all}} \in \mathbb{R}^{\text{nDOF} \times (n_{\text{snap.}} r_{\text{snap.}})}$$

3 Perform SVD

$$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* = \text{SVD}\left(\boldsymbol{\Psi}_{\text{all}}\right)$$

4 Select Glob. Modes, e.g.:

$$\boldsymbol{\Psi}_{\text{glob.}} = \boldsymbol{U}_{:,1:r_{\text{glob.}}}$$

$$\boldsymbol{\Psi}_{\text{glob.}} \in \mathbb{R}^{\text{nDOF} \times r_{\text{glob.}}}$$

5 formulate binding vector $\boldsymbol{g}$

6 Reduce MBS with $\boldsymbol{q} = \boldsymbol{\Psi}_{\text{glob.}} \boldsymbol{p}$

$$\boldsymbol{M}_{\text{red.}} = \boldsymbol{\Psi}_{\text{glob.}}^{\text{T}} \boldsymbol{M} \boldsymbol{\Psi}_{\text{glob.}}$$

$$\boldsymbol{K}_{\text{red.}} = \boldsymbol{\Psi}_{\text{glob.}}^{\text{T}} \boldsymbol{K} \boldsymbol{\Psi}_{\text{glob.}}$$

$$\boldsymbol{f}_{\text{red.}} = \boldsymbol{\Psi}_{\text{glob.}}^{\text{T}} \boldsymbol{f}$$

$$\boldsymbol{C}_{\text{q,red.}} = \boldsymbol{C}_{\text{q}} \boldsymbol{\Psi}_{\text{glob.}}$$

7 Lagrangians Type 1

$$\boldsymbol{g} = \boldsymbol{0}$$

$$\boldsymbol{M}_{\text{red.}} \ddot{\boldsymbol{p}} + \boldsymbol{K}_{\text{red.}} \boldsymbol{p} + \boldsymbol{C}_{\text{q,red.}}^{\text{T}} \boldsymbol{\lambda} = \boldsymbol{f}_{\text{red.}}$$